

Harshit Mittal



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Education

University of Leeds

Expected September 2026

Master of Science in Advanced Computer Science (First Class)

Leeds, United Kingdom

- **Relevant Coursework:** Algorithms (C++), Machine Learning (Python), Deep Learning (Python), Data Mining and Text Analysis (Python), Data Science (Python)
- **Thesis:** The design and implementation of SAPNeXt, a novel hierarchical encoder-decoder architecture integrating Spatial Adaptive Fusion and PixelShuffle upsampling over a pretrained ConvNeXt backbone, achieving a mean Dice of 0.8373 and AJI of 0.6307 on the MoNuSeg test set while reducing inference time to 1.6 seconds per slide, outperforming CellViT, HoVerNeXt, and SwinUNet across 19 tissue types on the PanNuke pan-cancer benchmark.

Experience

USIC&T

Jan 2025 – May 2025

AI Research Intern

Delhi, India

- Pioneered novel medical AI activation and class-balancing techniques, slashing information loss to 7% and boosting accuracy by up to 10% across peer-reviewed frameworks (Springer/IEEE).
- Architected containerized ML pipelines using Docker and Git, standardizing cross-team reproducibility and streamlining version-controlled feature deployments.

CFEES, DRDO

Aug 2023 – Oct 2023

Project Trainee

Delhi, India

- Engineered and deployed a peer-reviewed 3D facial recognition pipeline that upgraded CCTV footage to point-clouds, directly improving the department's active production security posture.
- Optimized cross-platform Flutter application performance to boost end-user satisfaction, establishing rigid code standards through meticulous Git history and documentation tracking.

Projects

AdaptAttend – Adaptive Attendance System | Flutter, Dart, TFLite, ML Kit, Drift, Next.js | [🔗](#)

- Built a fully offline Flutter attendance app supporting 3 methods – 6-digit TOTP, GPS geo-fence, and manager-side face scan – with zero backend dependency; all data stored on-device via Drift (SQLite).
- Integrated MobileFaceNet TFLite model (128-d embeddings, cosine similarity) with Google ML Kit face detection for on-device 1-to-N face recognition, replacing manual PIN entry and eliminating remote attendance fraud.
- Engineered a geo-fence security layer using Haversine distance to block attendance marking outside a configurable office radius, and restricted face scan to the manager's device to prevent spoofing.

ShopHandBook | Flutter, Dart, SQLite, pdf package, fl_chart, Provider | [🔗](#)

- Engineered a fully offline Android shop management app with 5 integrated modules – document scanner, PDF billing, inventory manager, customer tracker, and analytics – eliminating the need for internet connectivity or third-party cloud services.
- Implemented on-device multi-page PDF generation using the pdf package, enabling users to scan and merge multiple camera captures into a single invoice or document with zero server dependency.
- Built a local SQLite database layer via a singleton DatabaseService with full CRUD, relational bill-item joins, and dashboard aggregation queries, supporting complete shop data management entirely on-device.

Technical Skills

Languages: Python, Dart, C++, R, TypeScript, JavaScript, HTML, CSS

Technologies: Flutter, PyTorch, TensorFlow, Next.js, ML Kit, TFLite, Node.js, Docker, Git, SQLite, Drift

Concepts: Deep Learning, Computer Vision, Face Recognition, Transformers, Geo-fencing, On-device ML, Agile Methodology, CI/CD, Database Normalization, Cloud Computing